STUDY COMPANION · FALL 2026

English for Engineering I Weeks 5–7

All rules, questions and tasks from every course file for weeks 5–7 — the passive and describing processes, graph language and short talks, and noun formation for comparing solutions.

Compiled for independent study and Test 2 preparation
Tashkent University of Technology · Faculty of Energy Engineering
Sources: course Syllabus, Weeks 5–7 Lectures, Grammar Worksheet

Contents

1 Course Map and How to Use This Document 2

1.1 Weekly plan and sources (weeks 5–7) 2

2 Week 5 — Describing Processes and Sequences; the Passive 3

2.1 Grammar catalogue for Week 5 3

2.2 Week 5 Lecture, Part 1: Grammar — Passive — questions and answers 4

2.3 Week 5 Lecture, Part 2: listening and oral expression — some manufacturing processes . . . 7

2.4 Week 5 Lecture, Part 3: reading comprehension — «How Is Tea Made?» 8

2.5 Week 5 Lecture, Part 4: describing a process 10

2.6 Week 5 Grammar Worksheet — tasks and answers 12

3 Week 6 — Understanding Graphs and Short Talks 15

3.1 Vocabulary and speaking catalogue for Week 6 15

3.2 Week 6 Lecture — questions and answers 18

4 Week 7 — Comparing Solutions; Noun Formation 25

4.1 Grammar catalogue for Week 7 25

4.2 Week 7 Lecture — questions and answers 27

5 Appendix 30

5.1 Passive quick reference 30

5.2 Graph language bank 30

5.3 Noun formation table 31

5.4 Ten one-line answers 31

1 Course Map and How to Use This Document

This document is a complete, self-contained companion for weeks 5–7 of the Foundation course **English for Engineering – I** at Tashkent University of Technology (Fall 2026, 4 ECTS; module coordinators Ella Tsoy and Florence Dujarric, **Faculty of Energy Engineering**). The module is mandatory for all Bachelor of Science tracks — 60 classroom hours (30 lecture + 30 practical) plus 60 hours of independent study, with no prerequisites or co-requisites — and it prepares Foundation students to move from general school English to confident, accurate and organized everyday and academic communication. This companion was compiled by hand from every relevant course file for the block: the **Week 5, Week 6 and Week 7 lecture decks** and the **Week 5 grammar worksheet** (the passive and its uses). The Week 5–6 files are image scans transcribed by OCR, and obvious scanning artifacts have been cleaned, but every question, gap-fill, matching task, definition and discussion prompt that appears in those sources is reproduced word-for-word, and every exercise is followed by a full answer key. Under the syllabus assessment table, **Test 2**, which covers this block, is a paper-based writing task (1 hour, 10 %), so the grammar catalogues below receive the most detailed treatment.

The scope follows the official course structure. **Week 5** opens the block with the topic *Describing Processes and Sequences*: the grammar strand is the **passive** (all tenses), previewed through the TED talk *The solution to our energy problems… is a toaster*; the lecture then works through the manufacturing-process vocabulary (drilling, casting, welding…), the reading *How Is Tea Made?*, and a describing-a-process strand built around a solar-radiation text and its sequencing language. **Week 6** turns to *Understanding Graphs and Short Talks*: the oral-expression strand trains describing simple graphs and charts (diagram types, movement verbs and nouns, adjective and adverb pairs), the lecture adds cause-and-effect language and a listening strand on the structure and signposting of short talks, plus the TED talk *How to Spot a Misleading Graph*. **Week 7** closes the block with *Comparing Solutions*: the grammar strand is **noun formation** — introduced with the word formation video «Virtually Fluent» and worked through suffixes, prefixes, roots, compound nouns and conversion — followed by reading comprehension and the Switch Games oral expression.

Each weekly part of this document has the same internal structure. First comes the rules catalogue: every rule, form table and language box that the week's sources teach, written out formally and numbered continuously so that any rule can be cited from the answer keys. Then come the questions and tasks, grouped exactly as they appear in the source files, each tagged with its source and followed immediately by a complete answer key. Listening and video activities are reproduced with their tasks; where an answer depends on a recording or an image page that only students in class can see, the key marks the expected answer type and the most probable completion, so the exercise can still be checked. The appendix at the end collects quick-reference material — a passive quick reference, a graph language bank of verb + preposition patterns, a noun formation table and ten one-line revision answers — so that the question bank here is complete «to the end».

1.1 Weekly plan and sources (weeks 5–7)

[Source: course Syllabus; course files for weeks 5–7]

Week	Topic	Sources
5	Describing Processes and Sequences: grammar — the passive (all tenses); TED talk «The solution to our energy problems… is a toaster»; listening and oral expression — manufacturing processes; reading comprehension — «How Is Tea Made?»; describing a process (solar radiation; sequencing words)	e_{w5lec}.txt (Week 5 Lecture, OCR, 13 pages); e_{w5gw}.txt (Week 5 Grammar Worksheet — the passive and its uses, OCR, 5 pages)

Week	Topic	Sources
6	Understanding Graphs and Short Talks: oral expression — describing simple graphs and charts; TED talk «How to Spot a Misleading Graph»; cause-and-effect language; listening to short talks — structure and signposting	e_{w6lec}.txt (Week 6 Lecture, OCR, 14 pages)
7	Comparing Solutions: grammar — noun formation (word formation video «Virtually Fluent»; suffixes, prefixes, roots, compound nouns, conversion); reading comprehension; oral expression — Switch Games	$Week_{7Lecture}$.txt (Week 7 Lecture, text layer, 14 pages)

2 Week 5 — Describing Processes and Sequences; the Passive

2.1 Grammar catalogue for Week 5

[Source: Week 5 Lecture (e_{w5lec}.txt), Part 1 — Language Reference: passive (all tenses); Week 5 Grammar Worksheet (e_{w5gw}.txt), rules page «the passive and its uses»]

Week 5 has one grammar block: the passive, in all the tenses the course needs for describing processes and sequences. The rules below collect the language-reference panel of the Week 5 lecture and the complete rules page of the Week 5 grammar worksheet; every example is quoted from the two files.

- **Form.** We often use the passive (**be + past participle**) when it isn't said, known, or important who does an action: *A lot of films are shot on location. The director died when the film was being made. When is our car being repaired? You'll be picked up at the airport by one of our staff. This bill must be paid tomorrow. I love being given a massage. The new concert hall will be opened by the King.* In passive sentences, the object of the verb becomes the new subject: *Artem's bike has been stolen.* (= somebody has stolen Artem's bike, but we don't know who.)
- **Use of the passive (worksheet rules page).** Use the passive (a) to move important information — the object of the active sentence — to the beginning of the sentence: *Two men are being questioned by the police.*; (b) to be impersonal and not mention the name of the person who performs the action — *we often use the passive in scientific or technical processes: Oil and gas are found beneath the ground or sea*; (c) when it is obvious who performed the action (e.g. teachers), or when we do not want to use a general subject such as *people, they*: *Aztec boys were trained to fight. Two missing children have been found safe and well.*; (d) more often in formal speech and writing than in everyday speech.
- **The agent by.** If we want to say who did the action, we use **by** (lecture). Use **by** when it is important to know the «agent» or person who performed the action — *Some countries have been harmed by tourism. Many buildings in the city have been damaged by the earthquake.* — and only when it is really necessary: *The two men were arrested.* (by the police — this is obvious).
- **Active or passive — two ways of focusing.** We can often say things in two ways, in the active or in the passive. Compare: *Napoleon was directed by Ridley Scott.* (= we want to focus more on the film) / *Ridley Scott directed Napoleon.* (= we want to focus more on the director).
- **Negatives and questions** are formed in the same way as in active sentences: *Some films aren't shot on location. Is your car being repaired today?*
- **Present simple passive: am/is/are + past participle** (for processes that are generally true): *Oil and gas are found beneath the ground or sea. This room isn't used very often.* **Past simple passive: was/were + past participle** (for finished events): *The college was built in the 16th century. Aztec boys were trained to fight.*

- **Transitive and intransitive verbs.** Verbs which have an object are called transitive verbs; verbs without an object are intransitive. **Only transitive verbs can be made passive:** *I lost my wallet on the way to school.* (transitive) / *I swim every day.* (intransitive).
- **The preposition with.** Use **with** for objects which perform actions: *The game was played with a ball of solid rubber.*
- **The passive in science and engineering.** We often use the passive to talk about processes, because the steps of the process matter more than the person doing them: the manufacturing descriptions of the lecture are all passive — *A hole is made using a drill. A material is pressed into the correct shape. Hot liquid material is poured into a container and cooled…* — and the same pattern runs through the week's science text: *Around 25% is then reflected by the upper atmosphere and clouds… Finally, 50% is absorbed by the Earth's surface.*
- **Sequencing a process.** There are a number of different ways to talk about events in a process. You can use adverbs — **firstly, then, next, after that, finally** (*Firstly, some energy is removed by the clouds.*). You can also use words like **after, as, as soon as, once** and **when** to show how two events relate to each other (*As soon as sunlight hits the flower, its petals open.*). The phrase **at this stage** is used to help create a clear sequence (*The temperature reaches 100°C. At this stage, the water in the container begins to boil.*).

Tense	Active	Passive
present simple	We find oil and gas beneath the ground or sea.	Oil and gas are found beneath the ground or sea.
present continuous	The police are questioning two men.	Two men are being questioned by the police.
past simple	Teachers trained Aztec boys to fight.	Aztec boys were trained to fight.
present perfect	Tourism has harmed some countries.	Some countries have been harmed by tourism.
will	We will choose a new class representative on Monday.	A new class representative will be chosen on Monday.

2.2 Week 5 Lecture, Part 1: Grammar — Passive — questions and answers

[Source: Week 5 Lecture (e_{w5lec}.txt), pages 1–5 — introduction, TED talk preview, quick check, listening for gist, Language Reference, Activities 1–2]

2.2.1 Preview activities — TED talk «The solution to our energy problems… is a toaster»

Before listening, the lecture previews the key vocabulary of the talk in three panels. Learn each term together with its definition and the way it is used in the talk — the definitions are the answers.

Environmental & Energy Terms

Preview 1 — Fossil Fuels (noun). Give the definition and the example from the talk.

Answer. Definition: natural energy sources formed from the remains of ancient plants and animals (such as coal, oil, and natural gas). In this talk: most industrial manufacturers burn fossil fuels to create high-temperature heat.

Preview 2 — Carbon Pollution / Emissions (noun). Give the definition and the example from the talk.

Answer. Definition: greenhouse gases (mostly CO_2) released into the atmosphere that trap heat and damage the climate. In this talk: industrial heat is responsible for roughly 20% of global carbon pollution.

Preview 3 — Intermittent (adjective). Give the definition and the example from the talk.

Answer. Definition: not continuous; stopping and starting at irregular intervals. In this talk: renewable energy from wind and solar is intermittent because it depends on the weather (the wind blowing or the sun shining).

Materials & Mechanical Terms

Preview 4 — Composite (noun). Give the definition and the example from the talk.

Answer. Definition: a material made by combining two or more distinct elements to create a new substance with improved properties. In this talk: a nickel and chromium composite was invented in 1905 to make better heating elements.

Preview 5 — Filament / Heating Element (noun). Give the definition and the example from the talk.

Answer. Definition: a thin wire or metal piece that converts electricity into thermal energy (heat) due to electrical resistance. In this talk: the large storage bricks are threaded with filaments, just like the ones inside a kitchen toaster.

Preview 6 — Abundant (adjective). Give the definition and the example from the talk.

Answer. Definition: available in large quantities; plentiful and easy to find. In this talk: materials like bricks, rocks, and molten salt are cheap and abundant, making them great for storing heat.

Thermal & Structural Terms

Preview 7 — Disperse (verb). Give the definition and the example from the talk.

Answer. Definition: to scatter, spread out, or distribute something evenly over a wide area. In this talk: coiled toaster wires allow heat to be dispersed evenly across a slice of bread or a structural brick.

Preview 8 — Encased (adjective/verb). Give the definition and the example from the talk.

Answer. Definition: completely enclosed, covered, or wrapped in a protective container or outer layer. In this talk: the entire thermal battery system is encased in insulating steel.

Preview 9 — Insulation (noun). Give the definition and the example from the talk.

Answer. Definition: a material used to trap heat, electricity, or sound and prevent it from escaping. In this talk: steel insulation ensures that less than 1% of the heat escapes the battery chamber daily.

Preview 10 — Channels (noun). Give the definition and the example from the talk.

Answer. Definition: narrow, open pathways or passages through which air, gas, or liquids can flow. In this talk: the interlocking blocks are molded with open channels so that cold gases can be blown through them to collect the stored heat.

2.2.2 Quick Check: Preview Quiz

Before listening, fill in the blanks using words from the glossary above.

Quick Check 1. Solar power is highly dependent on the weather, making it an ___ energy source.

Answer. **intermittent** — not continuous; stopping and starting at irregular intervals (renewable energy from wind and solar is intermittent because it depends on the weather).

Quick Check 2. To keep the heat inside the system from escaping into the factory, the brick chamber must be heavily ___.

Answer. **insulated** — protected with insulation, a material used to trap heat and prevent it from escaping (steel insulation ensures that less than 1% of the heat escapes the battery chamber daily).

Quick Check 3. Concrete is a common building material made from a ___ of cement, sand, and gravel.

Answer. **composite** — a material made by combining two or more distinct elements to create a new substance with improved properties.

Quick Check 4. Cool air is forced through open ___ inside the hot bricks to absorb the thermal energy.

Answer. **channels** — narrow, open pathways or passages through which air, gas, or liquids can flow (the interlocking blocks are molded with open channels so that cold gases can be blown through them to collect the stored heat).

2.2.3 Listening for gist

Choose the best answer for each question.

Listening for gist 1. What is the main topic of this TED-Ed talk? A) The history of kitchen appliances and how they changed breakfast. B) Using a century-old toaster technology to build large-scale heat batteries for industry. C) The process of manufacturing high-quality bricks, rocks, and molten salt. D) How wind and solar energy can completely replace the electric grid today.

Answer. **B) Using a century-old toaster technology to build large-scale heat batteries for industry.** The talk explains how the heating element of a toaster — a nickel and chromium composite invented in 1905 — is now threaded through cheap, abundant storage bricks so that intermittent renewable energy can be stored as heat for industry.

Listening for gist 2. Why is industrial heat a major environmental problem according to the talk? A) It consumes over 80% of the world's fresh water supply. B) It is too expensive for small companies to generate safely. C) Most companies burn fossil fuels to achieve high temperatures, creating 20% of global carbon pollution. D) Industrial factories release too much hot air directly into the atmosphere, causing global warming.

Answer. **C) Most companies burn fossil fuels to achieve high temperatures, creating 20% of global carbon pollution.** The glossary line confirms it: industrial heat is responsible for roughly 20% of global carbon pollution.

2.2.4 Activity 1 — circle the correct form, active or passive

Circle the correct form, active or passive. The first item is given: *The college built / was built in the 16th century.* → **was built**.

- The costumes for the show are making / are being made by hand.
- The story inspired / was inspired him to make a film.
- This castle hasn't inhabited / hasn't been inhabited for nearly a century.
- His latest film set / is set in Vietnam in the 1960s.
- The film will shoot / will be shot in the autumn.
- The actors aren't recording / aren't being recorded the dialogue until next week.
- The house wasn't using / wasn't being used by the owners during the winter.
- The make-up artist has transformed / has been transformed the actor into a monster.
- They hadn't owned / hadn't been owned the company for very long before they went bankrupt.
- The photo took / was taken by my husband on the balcony of our hotel.

Activity 1 — answer key (10 items).

Answer. (1) are being made; (2) inspired — the story performs the action, so the active is correct; (3) hasn't been inhabited; (4) is set; (5) will be shot; (6) aren't recording — the actors perform the action (active); (7) wasn't being used; (8) has transformed — the make-up artist performs the action (active); (9) hadn't owned — active; (10) was taken. Rule of thumb: choose the passive when the subject undergoes the action and the doer is unknown or unimportant (Rules 1–2); keep the active when the

subject performs the action on an object (Rule 7).

2.2.5 Activity 2 — complete with the passive so that the meaning is the same

Complete with the passive so that the meaning is the same. Only use *by* if necessary. The first item is given: *People don't use this room very often.* → **This room isn't used very often.**

- They subtitle a lot of foreign films. → A lot of foreign films ___.
- Garcia Marquez wrote Love in the Time of Cholera in 1985. → Love in the Time of Cholera ___ in 1985.
- Someone is repairing my laptop at the moment. → My laptop ___ at the moment.
- They haven't released the DVD of the film yet. → The DVD of the film ___.
- They won't finish the film until the spring. → The film ___ until the spring.
- You have to collect the tickets from the box office. → The tickets ___ from the box office.
- They hadn't told the actor about the changes in the script. → The actor ___ about the changes in the script.
- Damien Chazelle directed La La Land. → La La Land ___.
- They've already recorded the soundtrack. → The soundtrack ___.
- They were interviewing the director about the film. → The director ___ about the film.

Activity 2 — answer key (10 items).
Answer. (1) A lot of foreign films are subtitled. (2) Love in the Time of Cholera was written in 1985. (3) My laptop is being repaired at the moment. (4) The DVD of the film hasn't been released yet. (5) The film won't be finished until the spring. (6) The tickets have to be collected from the box office. (7) The actor hadn't been told about the changes in the script. (8) La La Land was directed by Damien Chazelle. (9) The soundtrack has already been recorded. (10) The director was being interviewed about the film. Note the tenses: present simple (1), past simple (2, 8), present continuous (3), present perfect (4, 9), will (5), have to (6), past perfect (7), past continuous (10). *By* is only necessary in (8), where the director's name is the point of the sentence (Rule 3); in (2) the writer's name is unnecessary because the sentence focuses on the book.

2.3 Week 5 Lecture, Part 2: listening and oral expression — some manufacturing processes

[Source: Week 5 Lecture (e_{w5lec}.txt), pages 6–8]

2.3.1 Task 1 — match each picture with a description

Match each picture with a description. The ten pictures in the lecture are labelled with the process names (legible in the scan: bolting, casting, drilling, inspection, painting, pressing, punching, turning, welding; the tenth label belongs with description 5).

- A hole is made using a drill.
- A surface is covered with paint.
- A hole is punched in a material.
- The work is checked carefully.
- A surface is polished to make it smooth and shiny.
- A material is pressed into the correct shape.
- Hot liquid material is poured into a container and cooled to make the correct shape.
- Two pieces are joined together using very high temperature.
- A material is turned on a machine called a lathe. Part of the material is removed to make the correct shape.
- Two pieces are joined together using a nut and a bolt.

Task 1 — answer key (10 pictures).

Answer. 1 — drilling; 2 — painting; 3 — punching; 4 — inspection; 5 — polishing; 6 — pressing; 7 — casting; 8 — welding; 9 — turning; 10 — bolting. Notice that every process description is a passive sentence in the present simple (Rule 9).

2.3.2 Task 2 — a manufacturing process (check valve)

1 Look at the diagrams. Listen to the description of the manufacturing process for a check valve. Number the steps in the correct order. (The six diagrams of the lecture show bolting, drilling, turning, inspection, casting and painting.) — The numbering depends on the recording; from the diagrams and the descriptions the expected order follows the manufacturing logic: the body of the valve is cast, then turned on a lathe, then the holes are drilled, the parts are bolted together, the valve is painted, and finally the work is inspected.

2 Listen again. Answer the questions.

Task 2, question 1. What verb tense does the speaker use to explain the process?

Answer. The present simple passive — *is/are* + *past participle* (Rule 6): a hole is made, a surface is covered, the pieces are joined…

Task 2, question 2. Why is this tense used?

Answer. Because in a manufacturing process it is the steps that matter, not the person who performs them: who does the action is not said, known or important (Rule 1), and the passive is the standard way of describing scientific and technical processes (Rules 2 and 9).

Task 2, question 3. What words does the speaker use to explain the order of the steps in the process?

Answer. Sequencing words: first(ly), then, next, after that, finally — the adverbs of the sequencing box (Rule 10).

2.3.3 Speaking — explaining a manufacturing process

Work in pairs. Student A uses handout A, and student B uses handout B to explain a manufacturing process. (The handouts are distributed in class; the model below shows the expected style.)

Speaking task — model answer.

Answer. A full description uses the present simple passive plus sequencing words: *First, the body of the valve is cast from hot metal and cooled to make the correct shape. Then it is turned on a lathe, and part of the material is removed. Next, a hole is drilled, and the parts are bolted together with a nut and a bolt. After that, the valve is painted. Finally, the work is checked carefully — the valve is inspected.*

2.4 Week 5 Lecture, Part 3: reading comprehension — «How Is Tea Made?»

[Source: Week 5 Lecture (e_{w5lec}.txt), pages 8–10]

2.4.1 Activity 1: Vocabulary

Before reading, match each word from the article (1–10) with its meaning (A–J). Write the letter in the Answer column.

- 1 pluck (plucked)
- 2 moisture
- 3 brew
- 4 wither (withering)
- 5 oxidation
- 6 enzymes

- 7 granules
- 8 bruise
- 9 grade
- 10 tea chest

- A. very small, rough pieces, like tiny grains
- B. to make a hot drink by pouring hot water onto tea leaves
- C. a large wooden box used to store and transport tea
- D. a process in which something reacts with the air and changes colour
- E. to pick leaves or flowers from a plant
- F. to damage the surface of something by pressing or hitting it
- G. to lose water and become soft and weak
- H. natural chemicals in living things that cause changes
- I. the small amount of water inside something
- J. to put things in order by size, type or quality

Activity 1 — answer key (matching 1–10 to A–J).

Answer. 1 — E (to pick leaves or flowers from a plant); 2 — I (the small amount of water inside something); 3 — B (to make a hot drink by pouring hot water onto tea leaves); 4 — G (to lose water and become soft and weak); 5 — D (a process in which something reacts with the air and changes colour); 6 — H (natural chemicals in living things that cause changes); 7 — A (very small, rough pieces, like tiny grains); 8 — F (to damage the surface of something by pressing or hitting it); 9 — J (to put things in order by size, type or quality); 10 — C (a large wooden box used to store and transport tea).

Check in pairs: use two of the words to say what you think happens to tea leaves in a factory. You will check your ideas when you read. — Model: *First the leaves are plucked and taken to the factory, where they are withered — they lose their moisture and become soft and weak.*

2.4.2 Activity 2: True or false?

Read the article. Write T (true) or F (false) for each statement. If a statement is false, correct it. (The article pages are image-only in the course file; the key below gives the expected answers — items 1, 3, 4 and 8 are confirmed by the vocabulary definitions and by Activity 3.)

- Tea factories are usually far from the tea gardens.
- Only 24 hours pass between plucking the leaves and packing the tea.
- The withering stage takes only a few minutes.
- During oxidation, the leaves change colour from green to brown.
- In China, teas are always named after the person who made them.
- CTC tea was invented to increase the weight of tea that could be packed into a sack.
- Oolong tea is made from fully oxidised leaves.
- Green tea is made without the oxidation stage.

Activity 2 — answer key (8 statements).

Answer. (1) F — tea factories are usually built close to the tea gardens, so that the leaves can be processed quickly after plucking (see Activity 3, question 1). (2) T — the whole process from plucking to packing takes about 24 hours. (3) F — the withering stage takes several hours: the leaves lose their moisture and become soft and weak (definition G), not a few minutes. (4) T — during oxidation the leaves react with the air and change colour from green to brown (definition D). (5) F — teas are not always named after the person who made them; many are named after other things, such as the place where they are made. (6) T — CTC tea is crushed, torn and curled into small granules, so

more tea fits into a sack. (7) F — oolong tea is made from partially oxidised leaves. (8) T — green tea is made without the oxidation stage (see Activity 3, question 6).

2.4.3 Activity 3: Short answers

Answer each question in one or two full sentences, using your own words where you can.

Activity 3, question 1. Why are tea factories usually built close to the tea gardens?

Answer. Because the leaves must be processed very soon after plucking; if the factories were far away, the leaves would lose their moisture and quality on the way.

Activity 3, question 2. Why are the leaves separated into batches of the same size?

Answer. So that all the leaves in a batch are processed in the same way: leaves of the same size wither, roll and oxidize evenly, which gives a consistent, well-graded tea.

Activity 3, question 3. What are the four stages of the orthodox method?

Answer. Withering, rolling, oxidation and drying — after drying, the tea is graded and packed.

Activity 3, question 4. What happens to the leaves during oxidation, and what does this stage decide?

Answer. During oxidation the enzymes inside the leaves react with the air, and the leaves change colour from green to brown and develop their flavour. This stage decides the type and strength of the tea — how dark and strong it will be.

Activity 3, question 5. How is the CTC method different from the orthodox method of rolling?

Answer. In the orthodox method the leaves are rolled into twisted pieces, but in the CTC method the leaves are crushed, torn and curled by machines into small granules; the process is faster, and the granules pack more tea into a sack.

Activity 3, question 6. Why does green tea have a light, fresh flavour?

Answer. Because green tea is made without the oxidation stage: the leaves are heated so that the enzymes stop working, and the leaves keep their light, fresh flavour.

2.5 Week 5 Lecture, Part 4: describing a process

[Source: Week 5 Lecture (e_{w5lec}.txt), pages 11–13 — text A, glossary, usage note, Activities 1–2, texts B and C, tasks 3–6]

2.5.1 Text A — solar radiation (model text with glossary and usage note)

The model text of the strand (adapted from Johnson, K. (2016). GCSE Physics For You. (5th ed.). Oxford: Oxford University Press, p. 214): *All the planets absorb heat from the sun in a process known as solar radiation. However, for life on our planet to continue, Earth must stay at an average surface temperature of around 15°C. For the Earth to stay at a constant temperature, it needs to absorb and emit radiation at the same rates. As solar radiation travels to Earth, there is a series of steps which reduce the amount of radiation that reaches the surface: firstly, some is removed as it enters the atmosphere — the mixture of gases that surround our planet. Around 25% is then reflected by the upper atmosphere and clouds, while a further 20% is absorbed by them. 5% of the radiation is reflected from the surface of the Earth — by deserts or ice. Finally, 50% is absorbed by the Earth's surface. Around 10% of this heat radiates from the surface, and the remaining 40% moves to the clouds and atmosphere. This heat is then radiated back out into space, completing the cycle.*

Glossary: **absorb** (v): to take in and keep heat, light, or other forms of energy; **emit** (v): to send out light or heat; **reflect** (v): to send back light or heat from a surface (the third definition line is only partly legible in the scan).

Usage note — sequence in a process (verbatim): There are a number of different ways to talk about events in a process. You can use adverbs: firstly, then, next, after that, finally — *Firstly, some energy is*

removed by the clouds. You can also use words like after, as, as soon as, once and when to show how two events relate to each other — *As soon as sunlight hits the flower, its petals open.* The phrase at this stage is used to help create a clear sequence — *The temperature reaches 100°C. At this stage, the water in the container begins to boil.*

2.5.2 Activity 1 — sequencing words and phrases

Complete the student's summary of text A using sequencing words and phrases from the box: *as, at this stage, finally, firstly, then*.

Summary with gaps: «1 ___, solar radiation enters the Earth's atmosphere there are a number of steps that reduce it. 2 ___, approximately 25% is reflected by clouds and the upper atmosphere. 3 ___, a further 20% is absorbed by them. 4 ___, 5% is reflected back from the Earth's surface. 50% is absorbed by the Earth's surface. 10% of this heat radiates directly from the surface and 40% is transferred back to the upper atmosphere and clouds. 5 ___, this remaining amount radiates back into space.»

Activity 1 — answer key (5 gaps).

Answer. (1) As solar radiation enters the Earth's atmosphere, there are a number of steps that reduce it. (2) Firstly, approximately 25% is reflected by clouds and the upper atmosphere. (3) Then, a further 20% is absorbed by them. (4) At this stage, 5% is reflected back from the Earth's surface. (5) Finally, this remaining amount radiates back into space. Each word of the box is used once; *as* introduces the subordinate clause that relates the two events (Rule 10).

2.5.3 Activity 2 — process vocabulary

Complete the sentences using words from the box: *constant, process, removed, series, steps*.

- During the interview, he was asked a ___ of questions about his work experience.
- Child psychologists are interested in the ___ of learning among children.
- The number of students in higher education has remained ___.
- Following a large landslide, many trees and plants are ___ from the surface layer.
- The discovery of DNA was one of the first ___ in understanding how to cure genetic conditions.

Activity 2 — answer key (5 sentences).

Answer. (1) series; (2) process; (3) constant; (4) removed; (5) steps.

2.5.4 Texts B and C — important processes, tasks 3–6

Text B (verbatim): A blood culture involves taking a sample of blood from a patient and transferring the blood into a blood culture bottle. The bottle is then transported to the laboratory for processing. The procedure has three main stages: sample taking, transport and processing. — Glossary: **culture** (n): a group of cells grown in the laboratory for a medical or scientific study; **sample** (n): a small amount of a substance, often taken to be tested.

Text C (verbatim): Many companies who use internet-based questionnaires now include «progress indicators». These tell the person who is completing the questionnaire how many questions they have answered. They also often say how much more information needs to be given to complete the task. Previous research suggested that the indicators could have a negative effect on how many people answer questionnaires. However, people who regularly answer these kinds of questionnaires report that if they know how long the process will be, they will answer more honestly and completely. It is also important that the last question in the process allows respondents to give their own opinions.

Task 3. Read texts B and C. Match the content to the correct subject: 1 business studies ___ 2 medicine ___.

Answer. 1 business studies — text C (internet-based questionnaires and their progress indicators); 2 medicine — text B (the blood culture: sample taking, transport and processing).

Task 4. Choose the correct word to complete the definitions.

- last / previous — happening or existing before an event that you are talking about now
- stage / task — a piece of work that has to be done
- procedure / stage — a point, period or step in a process
- procedure / task — a series of actions done in a particular order or way
- previously / regularly — often
- involve / transfer — to move from one place to another

Task 4 — answer key (6 definitions).

Answer. (1) previous; (2) task; (3) stage; (4) procedure; (5) regularly; (6) transfer.

Task 5. Complete the sentences using the correct form of the highlighted words in texts B and C.

- To stay fit and healthy it is important to get ___ exercise.
- The final ___ of the experiment involved heating the materials to 1,000°C.
- The major ___ to the economy began with the banking crisis in 2008.
- The current energy minister has ___ worked for a major oil and gas company.
- The writing exam ___ describing a graph, and then writing an essay.
- The oesophagus ___ food from the mouth to the stomach.

Task 5 — answer key (6 sentences).

Answer. (1) regular (from *regularly*, text C); (2) stage (from *stages*, text B); (3) change (the position of the gap in this item is not fully legible in the scan — the sentence needs a noun for the major economic event that began in 2008); (4) previously (from *previous*, text C); (5) involved (from *involves*, text B); (6) transfers (from *transferring*, text B).

Task 6. Write a short paragraph (two or three sentences) about one of the topics below. Try to use some of the words from the unit: • Describe a process or cycle that you are familiar with. • Explain which processes are important in your area of study.

Answer. Model for the first topic (passive + sequencing words, Rules 9–10): *Bread is made in four stages. First, the grain is cleaned and ground into flour; then the flour is mixed with water and yeast, and the dough is left to rise. After that, the dough is shaped and baked at about 200°C, and finally the loaves are cooled and packed.* Model for the second topic: *In energy engineering, the most important processes are generation and distribution: first, fuel is burned in a furnace and water is heated to make steam; the steam is then used to drive a turbine, and the turbine drives a generator, so electricity is produced and carried to consumers through power lines.*

2.6 Week 5 Grammar Worksheet — tasks and answers

[Source: Week 5 Grammar Worksheet (e_{w5gw}.txt), exercises 1–6, pages 1–5; the rules page is reproduced in the catalogue above (Rules 1–8)]

2.6.1 Exercise 1 — past simple passive

Complete the sentence with the past simple passive form of the verb in brackets. Item a is given: *In ancient times, fires (light) were lit to use smoke as a signal over long distances.*

- b. Morse code (invent) ___ by an American called Samuel Morse.
- c. The world's first television images (show) ___ to scientists in Scotland in 1926.
- d. Radios (use) ___ to broadcast coded messages during the Second World War.
- e. The World Wide Web (design) ___ originally to make it easier for nuclear physics researchers to communicate.
- f. Buying books and CDs over the Internet (make) ___ popular by the company Amazon.

- g. The first mobile phones with cameras (sell) ___ in Japan.
- h. Vinyl albums and tapes (replace) ___ by CDs, which are now being replaced by mp3.
- i. The first English dictionary (wrote) ___ by Robert Cawdrey in 1604.
- j. The first mass-produced cars (introduce) ___ by the Ford company in the USA.

Exercise 1 — answer key (items b–j).

Answer. (b) was invented; (c) were shown; (d) were used; (e) was designed; (f) was made; (g) were sold; (h) were replaced; (i) was written — the worksheet gives the past form *wrote*, but the passive needs the past participle *written*; (j) were introduced. Every sentence describes a finished past event, so the form is was/were + past participle (Rule 6).

2.6.2 Exercise 2 — «How a toy car is made» (present simple passive)

Read the text and underline all the subject, verb and object structures. Then rewrite the text using present simple passive verbs. Leave out any unnecessary words. Item a is given: *As a first step, a computer is used to plan the exact shape of the car.*

The text (verbatim): *In the factory, they use a computer to plan the exact shape of the car. They then feed this computer program into a machine, and produce a plastic prototype. Then they produce the actual toy cars in a factory abroad. As far as construction is concerned, they make the bodies of the cars from plastic. They add small electric motors to the cars, and then they paint them. They also attach licence plates. Quality control inspects the cars and then they wrap them and pack them into cardboard boxes. They ship the cars to Britain.*

Exercise 2 — answer key (items a–g).

Answer. (a) As a first step, a computer is used to plan the exact shape of the car. (given) (b) This computer program is then fed into a machine, and a plastic prototype is produced. (c) The actual toy cars are produced in a factory abroad. (d) The bodies of the cars are made from plastic (as far as construction is concerned). (e) Small electric motors are added to the cars, and then they are painted; licence plates are also attached. (f) At the next stage, the cars are inspected by quality control, and then they are wrapped and packed into cardboard boxes. (g) Finally, the cars are shipped to Britain. The first words of items c–g are printed in the worksheet; item b, partly hidden in the scan, is reconstructed in the same pattern. The general subjects *they* are left out because they are unnecessary (Rule 2c).

2.6.3 Exercise 3 — «the Freedom Ship» (will passive or present perfect passive)

Complete the text with either the will passive or present perfect passive forms of the verbs in brackets. Item a is given: *In recent years many plans (make) have been made for large floating cities with living accommodation for as many as 50,000 people.*

The text (verbatim, with gaps): *One such project which b (advertise) ___ is the Freedom Ship. According to the plans for this project, this huge ship c (construct) ___ from smaller units, which d (take) ___ out to sea for final assembly. The ship e (power) ___ by 100 diesel engines, and the residents on board f (carry) ___ around the world continuously. The large number of people on the ship g (provide) ___ with all kinds of food, entertainment and sports facilities, and modern recycling methods h (use) ___ to save energy on board. At the time of writing, this project i (not build) ___, and j (criticize) ___ by some people who say it is too expensive. We shall have to wait and see whether this ambitious plan ever becomes reality.*

Exercise 3 — answer key (items b–j).

Answer. (b) has been advertised; (c) will be constructed; (d) will be taken; (e) will be powered; (f) will be carried; (g) will be provided; (h) will be used; (i) has not been built; (j) has been criticized. The plans that already exist at the time of writing take the present perfect passive; everything that will happen once the ship is built takes the will passive.

2.6.4 Exercise 4 — present perfect passive with by

Rewrite the sentence using a present perfect passive with by. Leave out any unnecessary words. Item a is given: *Cheaper air travel has made possible the global expansion in tourism. → The global expansion in tourism has been made possible by cheaper air travel.*

- b. Countries all over the world have experienced a growth in tourism.
- c. Millions of tourists have visited the popular Mediterranean resorts since the 1970s.
- d. Recently tourists have chosen more distant locations in Africa, Asia and South America.
- e. Mass tourism has badly affected some countries.
- f. International companies have built large hotels on unspoilt coastline.
- g. These developments have disturbed local wildlife.
- h. Tourists have damaged coral reefs and other habitats.
- i. Such developments have also displaced local people.
- j. Some governments have developed the idea of ecotourism to counteract some of these problems.

Exercise 4 — answer key (items b–j).

Answer. (b) A growth in tourism has been experienced by countries all over the world. (c) The popular Mediterranean resorts have been visited by millions of tourists since the 1970s. (d) More distant locations in Africa, Asia and South America have recently been chosen by tourists. (e) Some countries have been badly affected by mass tourism. (f) Large hotels have been built on unspoilt coastline by international companies. (g) Local wildlife has been disturbed by these developments. (h) Coral reefs and other habitats have been damaged by tourists. (i) Local people have also been displaced by such developments. (j) The idea of ecotourism has been developed by some governments to counteract some of these problems.

2.6.5 Exercise 5 — urban development (present perfect passive or past perfect passive)

Read the information about urban development. Then complete the text with the present perfect passive or past perfect passive form of the verbs in brackets. Item a is given: *In many countries in recent years, areas of urban land which were once used for industrial purposes (redevelop) have been redeveloped for other purposes.*

The text (verbatim, with gaps): *In London, for example, the Docklands b (transform) ___ since the 1980s. The docks c (build) ___ in the 19th century when London was a busy port, but by the 1980s, most business d (lose) ___ and many docks e (close) ___. This left large areas of derelict land, and many people without jobs. Since the 1980s, £10 billion ($17 billion) f (spend) ___ on the project. 600 hectares of derelict land g (improve) ___, 9 km of new roads h (build) ___, 80,000 new jobs i (create) ___ and 24,000 new homes j (construct) ___. In addition to this, 100,000 new trees k (plant) ___ and 130 hectares of parks l (create) ___. Cultural venues such as the Docklands Arena m (open) ___. Some local people do not agree with all the things that n (do) ___, as houses here are expensive, and most jobs are in media, IT or business. On the other hand, a declining area of the city o (give) ___.*

Exercise 5 — answer key (items b–o).

Answer. (b) has been transformed; (c) had been built; (d) had been lost; (e) had been closed; (f) has been spent; (g) has been improved; (h) have been built; (i) have been created; (j) have been constructed; (k) have been planted; (l) have been created; (m) have been opened; (n) has been done; (o) has been given. The docks were built, the business was lost and the docks were closed **before** the 1980s, so they take the past perfect passive (had been + past participle); everything since the 1980s takes the present perfect passive. The source page breaks off after o (give); the expected completion of the sentence is *has been given a new lease of life*.

2.6.6 Exercise 6A — rewriting without someone or people

Rewrite each sentence so that it does not contain the words *someone* or *people*. Item a is given: *Someone built ships with sails more than 5,000 years ago. → Ships with sails were built more than 5,000 years ago.*

- b. Someone constructed a hot-air balloon in 1783.
- c. People developed the steam ship in the 19th century.
- d. Someone opened the first successful passenger railway in 1830.
- e. People started the first passenger airlines after the First World War.

Exercise 6A — answer key (items b–e).

Answer. (b) A hot-air balloon was constructed in 1783. (c) The steam ship was developed in the 19th century. (d) The first successful passenger railway was opened in 1830. (e) The first passenger airlines were started after the First World War.

2.6.7 Exercise 6B — rewriting with the agent by

Rewrite each sentence so that it includes the agent *by*.

- f. The Chinese probably printed the first books more than a thousand years ago.
- g. The Remington company sold the first modern typewriters in the 1870s.
- h. Edison recorded the first words on a gramophone record in 1877.
- i. The Lumière brothers created the first modern cinema in France in 1895.
- j. Valdemar Poulsen invented the tape recorder in 1899.

Exercise 6B — answer key (items f–j).

Answer. (f) The first books were probably printed by the Chinese more than a thousand years ago. (g) The first modern typewriters were sold by the Remington company in the 1870s. (h) The first words on a gramophone record were recorded by Edison in 1877. (i) The first modern cinema was created by the Lumière brothers in France in 1895. (j) The tape recorder was invented by Valdemar Poulsen in 1899.

3 Week 6 — Understanding Graphs and Short Talks

3.1 Vocabulary and speaking catalogue for Week 6

[Source: Week 6 Lecture (e_{w6lec}.txt), Part 1 «Graphs and Diagrams» and «How to Describe Graphs and Charts in English», Part 2 «Cause-and-effect Language», Part 3 «Listening to Short Talks» with the Language Reference: Signposting Language]

3.1.1 Diagram types

- **Diagrams** are visual ways of presenting data concisely. They are often also called figures. In an academic article they are usually labelled Fig. (Figure) 1, Fig. 2, etc.
- **A pie chart** is a circle divided into segments from the middle (like slices of a cake) to show how the total is divided up. A key or legend shows what each segment represents.
- **A bar chart** is a diagram in which different amounts are represented by thin vertical or horizontal bars which have the same width but vary in height or length.
- **A histogram** is a kind of bar chart but the bar width also varies to indicate different values.
- **A table** is a grid with columns and rows of numbers.
- **A cross-section** is something, or a model of something, cut across the middle so that you can see the inside. A cross-section of the earth's crust, for example, shows the different layers that make it up. A label gives the name of each part of the cross-section. Cross-section can also be used to mean a small group that is representative of all the different types within the total group (e.g. the survey

looked at a cross-section of society).

- **A flowchart** is a diagram which indicates the stages of a process.
- **Common mistake.** Don't use the definite article (the) when referring to a specific diagram: *See Table 4* (NOT *See the Table 4*).

3.1.2 Graph language: axes, points and lines

- Graphs are drawn by **plotting points** on them and then drawing a line to join **adjacent** points. If there are two separate lines on a graph, the lines can cross or **intersect** at various points. Lines that run **parallel** to one another never intersect.
- Line charts and bar charts have a **vertical axis** and a **horizontal axis**; each axis is labelled to indicate what type of information it contains. Typical information included on the vertical and horizontal axis: age — how old; weight — how heavy; height — how tall; date — which day, month, year, etc.; time — how much time is required; length — how long; width — how wide; degrees — how hot or cold; percentage — a portion of 100%; number; duration — the length of time required.
- Model description (the lecture's pocket-money example): *The graph presents data relating to teenagers and pocket money. A random sample of 1,000 teenagers were surveyed and the average pocket money received at each age has been plotted on the graph. The x axis or horizontal axis indicates age and the y axis or vertical axis shows the amount of money received per week. The graph shows that 15-year-olds receive twice as much pocket money as 13-year-olds. From the graph we can see that the amount received reaches a peak at the age of 18 and then starts to decline. This decline can perhaps be explained by the fact that many teenagers start earning and stop receiving pocket money at the age of 18.*

3.1.3 Verbs and nouns of movement

- Graphs show how numbers **increase** or **decrease**. Numbers can also be said to **rise** or **grow** and **fall**, **drop** or **decline**. Other verbs used about growth include **double**, **soar**, **multiply**, **appreciate** and **exceed** [another number]: double = increase to twice the number or amount; opposite = halve; soar = increase very quickly and by a large amount; opposite = plummet; multiply = increase to a very large number; appreciate = increase in value; opposite = depreciate; exceed = increase to greater than a particular number or amount; opposite = fall below.
- **Language help.** The verbs increase and decrease are followed by **by** (e.g. *The population of the city has increased by 10%.*). The nouns increase, rise, growth, fall, drop and decline, decrease, are followed by **in** (to explain what is rising) or **of** (to explain the size of the change), e.g. *a rise / an increase of 15% in the number of cars*.
- **Positive movement** — verb and noun: to climb — a climb; to ascend — an ascent; to rise — a rise; to improve — an improvement; to recover — a recovery; to increase — an increase. Examples: *Sales have climbed over the past two quarters. We've experienced a rise in consumer demand. Consumer confidence recovered in the second quarter. There has been an increase of 23% since June. Have you seen any improvement in customer satisfaction?*
- **Negative movement** — verb and noun: to fall — a fall; to decline — a decline; to plunge — a plunge; to decrease — a decrease; to worsen — a slip; to deteriorate — a dip. Examples: *Research and development spending has fallen by 30% since January. Unfortunately, we've seen a decline over the past three months. As you can see, sales have plunged in the northwest region. Government spending has decreased by 10% over the past two years. There's been a slip in profits this past quarter. Comedy book sales have deteriorated for three quarters.*
- **Predicting future movement** — verb and noun: to project — a projection; to forecast — a forecast; to predict — a prediction. Examples: *We project improved sales in the coming months. As you can see from the chart, we forecast increased research and development spending next year. We predict improving sales through June.*

3.1.4 Adjectives and adverbs of change

Adjective — adverb	Meaning	Examples
slight — slightly	insignificant	There's been a slight decline in sales. / Sales have declined slightly over the past two months.
sharp — sharply	quick, large movement	Investment rose sharply during the first quarter. / We made a sharp increase in investment.
abrupt — abruptly	sudden change	Sales dropped abruptly in March. / There was an abrupt drop in sales in March.
rapid — rapidly	quick, very fast	We expanded rapidly throughout Canada. / The company made a rapid expansion throughout Canada.
sudden — suddenly	without warning	Unfortunately, consumer interest suddenly decreased. / There was a sudden decrease in consumer interest in January.
dramatic — dramatically	extreme, very big	We've dramatically improved customer satisfaction over the past six months. / As you can see from the chart, the dramatic growth has come after we invested in a new product line.
calm — calmly	evenly, without much change	The markets have reacted calmly to recent developments. / As you can see from the graph, consumers have been calm over the past few months.
flat	without change	Profit has been flat over the past two years.
steady — steadily	no change / level	There has been a steady improvement over the past three months. / Sales have improved steadily since March.

3.1.5 Cause and effect — verbs

Cause and effect is often described using conjunctions like because, prepositional expressions like due to and because of, and adverbs like therefore and consequently. This strand focuses on verbs and nouns relating to cause and effect — pay particular attention to the prepositions they are used with. You are probably already familiar with make, cause, create, do, produce and force; here are the other useful verbs of the lecture.

Verb	Example from the lecture	Meaning / comment
influence	Her grandmother influenced / had a considerable influence on Sarah's choice of career.	—
determine	Parental attitude largely determines how well a child adapts to school.	is the main factor affecting it
motivate	A good teacher motivates their students.	makes them want to do something positive
facilitate	The flow of traffic through the town is facilitated by the one-way system.	made easier
provoke	His speech provoked an angry response.	caused, usually something negative
trigger	The explosion was triggered by the heat.	started, usually something sudden and negative
account for	The tilting of the earth on its axis accounts for the change in the seasons.	explains
spring / stem from	Sid's determination springs / stems from his desire to improve the world.	is the result of
give rise to	The country's victory gave rise to a new mood in society.	—

Verb	Example from the lecture	Meaning / comment
contribute to	Mobile technology contributed to the information revolution.	was one factor influencing
stimulate	Their child was given drugs to stimulate growth.	cause something to develop or function
generate	Her exhibition generated a lot of interest.	caused something to happen or exist
induce	The drugs may induce nausea.	cause, often used in a medical context
inhibit	A teacher's presence can often inhibit teenagers' discussions.	prevent them from being as free as they might otherwise have been
derive from	A number of benefits can be derived from this situation.	gained as a result of something

3.1.6 Cause and effect — nouns

Noun	Example	Meaning / comment
chain reaction	The incident set off a chain reaction which affected us all.	set of related events in which each one causes the next one
effect, impact	The effect/impact of the film on the audience was very powerful.	note the preposition on
end, means	Does the end justify the means?	note how end here means aim or goal
origin, source	The accident was the origin/source of her later problems.	beginning or cause
precedent	There are several precedents for taking such a decision.	something that already happened and provides a reason for doing the same
reason	What was the reason for his success / why he succeeded?	note how reason is often followed by for or why

3.1.7 Short talk structure and signposting language

When listening to a lecture, you need to be able to recognize the main points and supporting details. It is also important to understand how information and ideas are organized. Lecturers may use signposting expressions to show how their lecture will be organized — recognizing these expressions will help you understand what you are going to hear and the order in which you will hear it. Lectures also often include definitions and explanations of key terms and ideas; it is important to recognize these in order to take effective notes. A short talk has three visible parts: the **introduction** (giving an overview), the **main points** (sequenced one by one) and the **conclusion**.

Function	Phrases
Giving an overview	As we'll see from today's lecture … / In today's lecture I'd like to look at … / In this lecture, I'll be focusing on …
Sequencing points	First, we'll look in more detail at … / Then, we'll move on to look at … / After that, we're going to see … / And then finally, I want to take …

Phrases for examples, explanations and referring back (Task 6 of the lecture): *So for example; An obvious example of this is* — giving an example; *this can be understood as; In other words; How that works is* — giving an explanation; *As I said; As we've already seen* — referring to something said earlier.

3.2 Week 6 Lecture — questions and answers

[Source: Week 6 Lecture (e_{w6lec}.txt), Practice Activities 1–4, the Statista practice activity, the TED talk questions, Part 2 Activities 1–5, Part 3 Tasks 1–6]

3.2.1 Part 1 practice — Activity 1: describe the chart «Cars entering downtown West City»

Look at the chart and complete the text with the missing words. (The chart shows the number of cars entering the downtown area of West City each day over an eight-year period, years 1–8; the totals rise for the first few years and then fall after year 6.)

The text with gaps: *The chart 1___ the number of cars entering the downtown area of West City each day over an eight-year period (years 1–8). The totals are listed on the 2___ axis (give two answers), while the years are listed on the 3___ axis. From the graph we see 4___. The number of cars 5___ for the first few years and 6___. The decline can be 7___ by the 8___ that a new mass transit railway was opened in year 6, which is a clear illustration of how good public transport can dramatically affect car use.*

Activity 1 — answer key (this page of the scan is partly unreadable; the key gives the expected completions).

Answer. (1) shows; (2) vertical / y (two answers are required); (3) horizontal / x; (4) the general trend — car use rose and then fell over the period; (5) rose / increased; (6) then fell / declined; (7) explained; (8) fact. The description follows the pocket-money model: what the graph presents, what the axes show, then the trend and its explanation.

3.2.2 Activity 2 — answer the questions

Activity 2, question 1. Draw examples of a pie chart and a bar chart.

Answer. A pie chart is drawn as a circle divided into segments from the middle (like slices of a cake), with a key or legend showing what each segment represents; a bar chart is drawn as thin vertical or horizontal bars of the same width whose height or length varies with the amounts.

Activity 2, question 2. What is the best type of diagram to present the different layers of rock in the Grand Canyon?

Answer. A cross-section — something cut across the middle so that you can see the inside; a cross-section of the rock shows the different layers that make it up, with a label giving the name of each part.

Activity 2, question 3. In a table, what is the difference between columns and rows?

Answer. A table is a grid with columns and rows of numbers: the columns run vertically (top to bottom) and the rows run horizontally (across the table).

Activity 2, question 4. What would be the best type of diagram to present the different stages in a research project you did?

Answer. A flowchart — a diagram which indicates the stages of a process.

Activity 2, question 5. How many segments are there in the pie chart in A opposite?

Answer. This item depends on the printed diagram: count the segments of the pie chart shown in Part 1 of the lecture — each segment («slice of the cake») represents one part of the total, labelled in the key or legend.

Activity 2, question 6. If you look at two adjacent columns in a table, are they next to each other or separated?

Answer. They are next to each other — adjacent means directly beside one another (as in «a line to join adjacent points»).

Activity 2, question 7. What is another name for a legend in a diagram?

Answer. A key («a key or legend shows what each segment represents»).

Activity 2, question 8. What type of data collection are you doing if you survey the first 50 people you come across?

Answer. You are taking a sample — but not the random sample of the pocket-money study: choosing the first people you come across does not give a representative cross-section of the population.

Activity 2, question 9. What do two lines on a graph do if (a) they intersect and (b) they run parallel to each other?

Answer. (a) They cross each other at a point; (b) they never intersect — lines that run parallel to one another never cross.

Activity 2, question 10. Choose the correct sentence: (a) There was an increase in 12% of smart phone sales last year. (b) There was an increase of 12% in smart phone sales last year.

Answer. (b) is correct: the noun increase takes **of** for the size of the change and **in** for what is rising — *an increase of 12% in smart phone sales.*

3.2.3 Activity 3 — replace the underlined words with more precise, academic words

- The different bits of the pie chart show the numbers of people in each age group.
- She kept a record by marking the midday temperature on a graph for a month.
- People's salaries usually reach their highest point when they are in their late 40s.
- This flowchart shows the different bits of our project over the next five years.
- The two lines on the graph cross each other at point A.
- Draw a line connecting the points that are next to each other.
- The government's popularity in the opinion polls is beginning to go down.
- If you look along the third line of the table you can see the figures for the 1950s.

Activity 3 — answer key (8 sentences).

Answer. (1) segments; (2) plotting (she kept a record by plotting the midday temperature on a graph); (3) reach a peak (salaries reach their peak in the late 40s); (4) stages; (5) intersect; (6) adjacent (the points that are next to each other); (7) decline (is beginning to decline); (8) row (look along the third row of the table).

3.2.4 Activity 4 — rewrite using the movement verbs; find the deliberate mistake

Rewrite the underlined words and phrases using words from B opposite (the movement panel). There is also a deliberate mistake in one of the sentences. Can you find it and correct it?

- Populations of some bird species in South Asia have crashed by 97% in recent years. The number of cases of death by poisoning has increased sharply.
- In 2007 the child mortality rate fell to lower than 60 deaths per 1,000.
- The average family car in the UK goes down in value by 20% per year. This means its value has fallen by more than half after just three years.
- A typical piece of land on the edge of the city will go up in value by 15% per year, and house prices have gone up rapidly by a large amount in the last six months.
- Business courses have increased greatly in number while science programmes have gone down.
- The temperature rose higher than 45°C in some parts of the country. See the figure 3.
- Between 1983 and 2006, the number of this species of eagle went up from 22 pairs to 58. Other bird populations have gone up to twice the number in the same period.
- The numbers of old soldiers attending regimental reunions are becoming smaller each year.

Activity 4 — answer key (8 sentences + the deliberate mistake).

Answer. (1) have plummeted by 97% (opposite of soar); the number of cases has soared; (2) fell below 60 deaths per 1,000; (3) depreciates by 20% per year (goes down in value); its value has more than

halved after just three years; (4) appreciates by 15% per year (goes up in value); house prices have soared in the last six months; (5) business courses have multiplied while science programmes have declined; (6) the temperature exceeded 45°C — and the deliberate mistake: *See the figure 3* must be corrected to *See Figure 3*, with no definite article (the common-mistake rule); (7) rose from 22 pairs to 58; other bird populations have doubled (gone up to twice the number); (8) are declining / decreasing each year (becoming smaller).

3.2.5 Practice activity — describing a graph from statista.com

Student A and student B — go to https://www.statista.com/, choose a graph, and describe it using the vocabulary from this section.

Statista practice activity — model answer.

Answer. A full description follows the pocket-money model: *The line graph presents data relating to passenger numbers on the city metro. The y axis or vertical axis shows the number of passengers per month, and the x axis or horizontal axis indicates the months of the year. From the graph we can see that passenger numbers rose steadily from January to June and reached a peak in July. After that, they declined slightly, and by December they had fallen to the January level. This decline can be explained by the fact that the universities were on holiday.*

3.2.6 TED Talk: How to Spot a Misleading Graph — listening comprehension

Listening for gist — answer the questions on the talk.

TED talk, question 1. What is the main topic of the video? a) How to make graphs using computer software. b) How graphs can mislead people, and how to spot it. c) Why politicians and advertisers should not use numbers. d) The history of statistics in advertising.

Answer. **b) How graphs can mislead people, and how to spot it.**

TED talk, question 2. What does the speaker say about the way people usually think of graphs? a) They are too complicated for most people. b) They are always accurate because they show hard numbers. c) People tend to trust them more than a plain claim, even though they can be manipulated. d) They are only used in science, not in advertising.

Answer. **c) People tend to trust them more than a plain claim, even though they can be manipulated.**

TED talk, question 3. What is the speaker's overall advice at the end? a) Never trust any graph. b) Only trust graphs from official organizations. c) Look carefully at the labels, numbers, scale, and context, and ask what story the graph is telling. d) Make your own graphs instead of using other people's.

Answer. **c) Look carefully at the labels, numbers, scale, and context, and ask what story the graph is telling.**

TED talk, question 4. In your own words, name two ways a graph can be misleading. (Open answer)

Answer. Model answers from the talk: (1) the vertical axis can be zoomed in or cropped so that a small difference looks dramatic; (2) the labels, numbers, scale or context can be manipulated or left out — for example, the baseline may be missing — so always ask what story the graph is telling.

3.2.7 Part 2: Cause-and-effect language — Activity 1

Choose the best verb to complete each sentence.

- Researchers are investigating why chocolate provokes / induces headaches in certain people.
- Wilson's most recent paper has motivated / generated a great deal of interest among sociologists.
- The drug triggered / facilitated headaches and dizziness among a number of subjects in the test.

- Having an end-of-term prize contributes / motivates students to do well in their class tests.
- Intensive farming has contributed to / provoked the decline of bird populations.
- The missile test stimulated / provoked an immediate and very strong response from neighbouring governments and from the United Nations.
- Astrologers believe that people's lives are motivated / determined by the planets and stars.
- The barrier inhibits / induces the flow of water into the area to prevent flooding in the rainy season.
- A leak in the tank stemmed from / accounted for 40% of the fuel loss, and evaporation took away another 5%.
- The renovated college buildings have wider doors to influence / facilitate the use of wheelchairs.
- The Minister cut taxes in an attempt to stimulate / trigger the economy, which was performing poorly.
- Britain has facilitated / derived many economic benefits from membership of the European Union.

Part 2, Activity 1 — answer key (12 sentences).

Answer. (1) induces (cause, often used in a medical context); (2) generated (caused something to happen or exist); (3) triggered (started, usually something sudden and negative); (4) motivates (makes them want to do something positive); (5) contributed to (was one factor influencing); (6) provoked (caused, usually something negative); (7) determined (is the main factor affecting it); (8) inhibits (prevents them from being as free as they might otherwise have been); (9) accounted for (explains); (10) facilitate (made easier); (11) stimulate (cause something to develop or function); (12) derived (gained as a result of something).

3.2.8 Activity 2 — match the beginnings and endings of the sentences

- 1 The article explores the origins — a major consequences for larger families.
- 2 One small explosion set off a chain — b precedents for his decision.
- 3 The confusion probably stemmed — c rise to a widespread sense of disillusionment.
- 4 The proposed new tax could have — d of the concept of democracy.
- 5 The disastrous events of 2003 gave — e reaction, causing massive damage.
- 6 The judge explained that there were — f from a lack of communication.

Activity 2 — answer key (6 pairs).

Answer. 1 — d (The article explores the origins of the concept of democracy); 2 — e (One small explosion set off a chain reaction, causing massive damage); 3 — f (The confusion probably stemmed from a lack of communication); 4 — a (The proposed new tax could have major consequences for larger families); 5 — c (The disastrous events of 2003 gave rise to a widespread sense of disillusionment); 6 — b (The judge explained that there were precedents for his decision).

3.2.9 Activity 3 — complete the sentences with the missing words

Complete the sentences with the missing words. In one sentence, it is not necessary to add a word.

- De Routa's work in the 1970s influenced ___ the development of computer science.
- The acid seemed to have no ___ whatsoever on the plastic, which remained unchanged.
- Everyone hoped that the outcome ___ the meeting between the two governments would be a new and better trade agreement.
- The reason ___ the failure of the project was a lack of funding.
- The impact of global warming ___ the polar ice caps is now understood more clearly.
- There is no precedent ___ this kind of decision.

Activity 3 — answer key (6 sentences).

Answer. (1) no word is necessary — influenced takes a direct object (Her grandmother influenced Sarah's choice of career); (2) effect (have no effect whatsoever on the plastic); (3) of (the outcome of the meeting); (4) for (the reason for the failure of the project); (5) on (the impact of global warming on the polar ice caps); (6) for (there is no precedent for this kind of decision).

3.2.10 Activity 4 — write the nouns related to these verbs

Activity 4. Write the nouns related to these verbs. Use a dictionary if necessary: motivate, influence, trigger, contribute, induce.

Answer. motivate → motivation; influence → influence (the noun has the same form); trigger → trigger (the noun has the same form); contribute → contribution; induce → inducement.

3.2.11 Activity 5 — complete the sentences using nouns from Activity 4

- Byron's poetry had a significant ___ on the work of other Romantic poets.
- Tree pollen can be a ___ for hay fever attacks in vulnerable people.
- All the political parties illegally offered people ___ to vote for them, such as cash payments, free tickets for sports events, etc.
- The ___ made by the burning of fossil fuels to global warming is now beyond dispute.
- The ___ to get a good job and to travel and see the world is often more effective than the teacher or the teaching material in language learning.

Activity 5 — answer key (5 sentences).

Answer. (1) influence (had a significant influence on the work); (2) trigger (can be a trigger for hay fever attacks); (3) inducements (offered people inducements to vote for them); (4) contribution (the contribution made by the burning of fossil fuels to global warming); (5) motivation (the motivation to get a good job and to travel).

3.2.12 Part 3: Listening to short talks — Task 1: using your knowledge to prepare for a lecture

1 Work in pairs. Discuss how the following factors are linked to motivation at work: praise from your boss, earning a lot of money, your job title, being creative, having power, doing challenging work, respect from colleagues, a good pension, a sense of achievement. 2 Which factors in 1 do you think are the most motivating?

Task 1 — model discussion answer.

Answer. Model answers: money matters most for people who have to support a family — this is Taylor's idea that money is the key motivator in the workplace; but praise from your boss, respect from colleagues and a sense of achievement often motivate people more than a higher salary, especially in creative or challenging work; a good pension gives security, which becomes more important with age. These are exactly the factors that the lecture's slides A–C connect to the theories of Taylor, Maslow, McGregor, and McClelland & Burnham.

3.2.13 Task 2: understanding the main points of a lecture

1 You are going to listen to extracts from a lecture about motivation at work. Look at slides A–C from the start of the lecture and decide which order they will appear in. Slide A: 1 What motivates people? 2 How have theories developed? 3 How have they influenced each other? Slide B: Theories of motivation — Taylor; Maslow; McGregor; McClelland & Burnham. Slide C: The history of motivational theories at work, Dr David Hughes, Institute of Psychology.

Task 2, question 1 — answer (slide order).

Answer. The most probable order is C → A → B: first the title slide of the lecture (C — The history of motivational theories at work, Dr David Hughes, Institute of Psychology), then the overview slide with the lecture's guiding questions (A — What motivates people? How have theories developed? How

have they influenced each other?), then the content slide listing the four theories (B — Taylor, Maslow, McGregor, McClelland & Burnham).

Task 2, question 2. Watch Extract 1 and check your prediction in 1. Make brief notes on the main points the lecture will cover.

Answer. Expected notes: the lecture will look at a number of theories related to motivation at work — first Taylor's idea that money is the key motivator; then one of the most famous theories of motivation, Maslow's hierarchy of needs; then how Maslow's ideas influenced the work of Douglas McGregor; and finally a slightly different view of motivation developed by McClelland and Burnham. (These are exactly the points that the signposting sentences of Task 3 announce.)

3 Work in pairs. Use your notes to tell each other what you understood from Extract 1.

3.2.14 Task 3: understanding the organization of a lecture

Watch Extract 1 again and complete the expressions in sentences 1–5 that show how the lecture will be organized.

- 1 ___, I'd like to look at a number of theories related to motivation …
- 2 ___ we'll look at Taylor's idea that money is the key motivator in the workplace.
- 3 Then, we'll ___ look at one of the most famous theories of motivation, which is Maslow's hierarchy of needs.
- 4 ___, we'll see how Maslow's ideas influenced the work of Douglas McGregor …
- 5 And ___, I want to ___ a slightly different view of motivation developed by McClelland and Burnham. …

Task 3 — answer key (5 signposting gaps).

Answer. (1) In today's lecture (giving an overview — the language reference: In today's lecture I'd like to look at …); (2) First (sequencing points — First, we'll look in more detail at …); (3) move on to (Then, we'll move on to look at …); (4) After that (After that, we're going to see …); (5) then finally … take (And then finally, I want to take …).

3.2.15 Task 4: taking notes on key information

1 Watch Extract 2 and label the timeline with the people behind the theories of motivation. (The people to place are: 1 McClelland and Burnham, 2 Taylor, 3 Maslow, 4 McGregor.)

Task 4, question 1 — answer (timeline).

Answer. The timeline runs chronologically: Taylor first (the earliest theory — money as the key motivator), then Maslow (the hierarchy of needs, which developed the field), then McGregor (whose work was influenced by Maslow's ideas), and finally McClelland and Burnham (the most recent, slightly different view).

2 Watch Extract 2 again and complete the notes: 1 Taylor: ___ motivates everyone. 2 Maslow: motivation = ___ main areas. 3 McGregor: management styles = ___ groups? 4 McClelland and Burnham: studied ___, main motivator = ___.

Task 4, question 2 — answer (expected completions; the notes depend on the recording).

Answer. (1) money — Taylor's idea is that money is the key motivator in the workplace, so money motivates everyone; (2) five — Maslow's hierarchy of needs has five levels; (3) two — McGregor divided management styles into two groups; (4) studied managers, and the main motivator = power.

3.2.16 Task 5: taking detailed notes on explanations and examples

1 Watch Extract 3 of the lecture. Complete the notes on each level of Maslow's hierarchy of needs. (The note table has two columns — Explanation and Example related to work; the rows printed in the lecture include Belonging and Security.)

Task 5, question 1 — expected completions of the note table.

Answer. Belonging — explanation: the need to feel part of a group and accepted by others; example related to work: good relationships with colleagues, being part of a team. Security — explanation: the need for safety, stability and protection; example related to work: job security and a stable salary, a safe working environment. The remaining levels of the hierarchy (physiological needs, esteem, self-actualization) follow the same pattern of explanation plus a work-related example.

2 Watch the last part of Extract 3 again and answer questions 1–4.

Task 5, question 2.1. Which need is fulfilled first?

Answer. The needs at the bottom of the pyramid — the basic, physiological needs — are fulfilled first.

Task 5, question 2.2. Can you move from security needs straight to esteem needs?

Answer. No — the levels are fulfilled in order, and the belonging (social) needs come between security and esteem, so they must be satisfied first.

Task 5, question 2.3. What happens when one need is met?

Answer. The next level up becomes the main motivator — the person moves up the pyramid.

Task 5, question 2.4. Does everyone have to reach the top of the pyramid to be satisfied?

Answer. No — not everyone reaches self-actualization, the top of the pyramid; people can be satisfied at different levels.

3 Compare your answers in 1 and 2 with another student. Help each other to complete any missing information.

3.2.17 Task 6: recognizing phrases for signposting a lecture

1 Match the following phrases the lecturer uses with functions a–c: As I said; So for example; this can be understood as; In other words; An obvious example of this is; How that works is; As we've already seen. (a giving an example, b giving an explanation, c referring to something said earlier.)

Task 6, question 1 — answer key.

Answer. a (giving an example): So for example; An obvious example of this is. b (giving an explanation): this can be understood as; In other words; How that works is. c (referring to something said earlier): As I said; As we've already seen.

2 Watch Extract 3 again and check your answers.

4 Week 7 — Comparing Solutions; Noun Formation

4.1 Grammar catalogue for Week 7

[Source: Week 7 Lecture (Week$_{7Lecture}$.txt), Part 1: Grammar — Noun formation; word formation video «Virtually Fluent» (https://www.youtube.com/watch?v=s7BtP$_{7lWPU}$)]

Week 7 has one grammar block: noun formation and the other word formation techniques used for comparing solutions. The lecture works through five panels — suffixes, prefixes, roots, compound nouns and conversion — with the video «Word Formation Techniques». The suffix, prefix, root and compound-noun panels of the course file are image pages; their rules are summarized below together with the fully transcribed conversion panel.

4.1.1 Word formation techniques — overview

- **Suffixes.** A suffix is added to the end of a verb or an adjective to form a noun. The noun-forming suffixes of the lecture's panel: **-tion** (act → action; inspect → inspection), **-ment** (develop →

development; measure → measurement), **-ness** (happy → happiness; dark → darkness), **-ity** (able → ability; possible → possibility), **-er** (build → builder; design → designer), **-or** (act → actor; generate → generator).

- **Prefixes.** A prefix is added to the beginning of a word and changes its meaning — most often by making a negative or a reversal. The lecture's prefix table is an image page; its practice (Activity 6) asks you to think of two more examples for each prefix in the table.
- **Roots.** With the roots method we add pieces to a root: **inspect → inspector → inspection**. A root carries the core meaning of the word family, and affixes are added around it.
- **Compound nouns.** A compound noun is a combination of two nouns written as one expression; the lecture builds word networks of compound expressions and practises them through network-completion, definition-guessing and collection tasks.
- **Conversion.** See the complete rule set below — conversion means that a word changes its job without changing its shape.

4.1.2 Conversion — complete rules (verbatim from the lecture)

What is conversion? Conversion means that a word changes its job without changing its shape. We add no prefix and no suffix. The same word simply starts to work as a different part of speech. • water (noun): *Can I have a glass of water?* • water (verb): *Please water the plants.* Compare this with the roots method. There, we add pieces to a root: inspect → inspector → inspection. In conversion, we add nothing: an email → to email. Teachers sometimes call this «zero derivation», because the ending is zero. Conversion is very common in English. It is one reason why English learners can use new words quickly: if you know the noun, you can often guess the verb.

Common patterns. The most common change is from a noun to a verb. Verb to noun is also very frequent.

Pattern	Word	Before	After
noun → verb	email	I got an email.	Email me tomorrow.
noun → verb	phone	My phone is new.	I'll phone you tonight.
noun → verb	book	I'm reading a book.	We booked a table.
noun → verb	brush	Use a brush.	Brush your teeth.
verb → noun	run	I run every day.	I went for a run.
verb → noun	drink	I drink tea.	Would you like a drink?
verb → noun	smile	She smiled at me.	She has a lovely smile.
verb → noun	help	Can you help me?	Thanks for your help.
adjective → verb	clean	The room is clean.	Please clean the room.
adjective → verb	empty	The box is empty.	Empty the box.
adjective → verb	calm	He stayed calm.	Calm down!
adjective → noun	the rich, the young	a rich man	The rich do not always feel happy.

With **the + adjective**, the word means a group of people: the poor, the old, the young, the unemployed. It takes a plural verb: *The young are often good with technology.*

How to tell which word you have. Because the shape does not change, the sentence tells you the job of the word. Ask three questions: 1. Where is it? After a, the, my, some or an adjective, it is a noun: *a drink, my phone, some help.* After a subject or after to, will, can, must, it is a verb: *I drink tea. I will phone you.* 2. Can it change form? A verb can take -s, -ed, -ing: *she emails, she emailed, she is emailing.*

A noun takes a/an or a plural -s: *an email, two emails.* 3. Does the stress move? A small group of two-syllable words change stress. The noun has stress on the first syllable, and the verb on the second.

Noun	Verb
a REcord	to reCORD
a PREsent	to preSENT
a CONtest	to conTEST
PROduce (fruit and vegetables)	to proDUCE (make)

A few words also change the final sound: a use /juːs/ → to use /juːz/; a house /haʊs/ → to house /haʊz/.

Watch out. • The meaning changes a little. When a noun becomes a verb, the new meaning is usually one of three types: put in or on (to butter bread = put butter on it), use (to phone someone = use a phone), or act as (to chair a meeting = be the chair). • Not every word converts. We say to email but not to computer. There is no rule. When you meet a new word, check a dictionary: it lists every job the word can do. • New conversions appear all the time: to google (search online), to zoom (have a video call), to ghost (stop answering someone), to bookmark (save a page). Many are informal, so check the style before you use them in an essay. • Do not confuse conversion with suffixes. Advice (noun) and advise (verb) are different words with different spelling. That is not conversion.

4.2 Week 7 Lecture — questions and answers

[Source: Week 7 Lecture (Week$_{7Lecture}$.txt), practice activities of the word formation panels, Parts 1–3]

Note on this file: the practice pages of the suffix, prefix, root and compound-noun panels are image-based in the course file, so only their readable instructions are transcribed below, with the expected way of working; the conversion activities are fully readable and are given with complete answer keys.

4.2.1 Suffixes — practice activities

Practice Activities: Activities 1–7 (suffixes). The seven activity pages are image-based in the course file and their sentences are not machine-readable; they practise the suffix panel reproduced in the catalogue (the first rule of the overview).

Suffixes — practice key.

Answer. Since the printed items are not readable in the scan, check every formation against the catalogue table: act → action, inspect → inspection, produce → production (-tion); develop → development, measure → measurement (-ment); happy → happiness, dark → darkness (-ness); able → ability, possible → possibility (-ity); build → builder, design → designer (-er); act → actor, generate → generator (-or). Watch the spelling changes (happy → happiness, possible → possibility).

4.2.2 Prefixes — practice activities

Practice Activities: Activities 1–5 (prefixes). These five activity pages are also image-based in the course file. Activity 6 is readable: *Think of two more examples for each prefix in the table.*

Prefixes, Activity 6. Think of two more examples for each prefix in the table.

Answer. The prefix table itself is an image page; practising with the commonest English prefixes, two more examples for each are: un- (unhappy, unlock), re- (rewrite, reuse), dis- (disagree, disconnect), in-/im- (incorrect, impossible), pre- (preview, preheat). Add your own examples for every prefix printed in the lecture's table.

4.2.3 Roots — practice activities

Practice Activities: Activities 1–3 and 5 (roots) are image-based in the course file. Activity 4 is readable: *Think of three other words based on each of the roots listed in the language reference. Put each into an appropriate phrase.*

Roots, Activity 4. Think of three other words based on each of the roots listed in the language reference. Put each into an appropriate phrase.

Answer. Model with the lecture's own root (inspect → inspector → inspection): three other words from the same root are spectator, respect and spectacular — *The stadium holds 40,000 spectators; He was always respectful to his teachers; The view from the summit was spectacular.* Repeat the pattern for every root in the lecture's reference table.

4.2.4 Compound nouns — practice activities

Compound nouns — combinations of two nouns. The language reference of this panel (word networks of compound expressions) is an image page in the course file.

Compound nouns, Activity 1. Complete these networks with any appropriate expressions from the language reference. Add extra bubbles if you need them.

Answer. This activity works with the printed networks: copy each network from the lecture, write the compound expressions from the language reference into the bubbles they belong to (an expression that combines with the central noun goes on its branch), and add extra bubbles for your own expressions.

Compound nouns, Activity 3. What are they talking about? In each case the answer is a compound noun from the language reference.

Answer. The definition sentences are printed as an image page in the course file; the task is solved by matching each definition to the compound noun from the reference whose two nouns together name the thing described — exactly the guessing game that Activity 4 then asks you to imitate.

Compound nouns, Activity 4. Now make up some sentences like those in Activity 3 relating to some of the new expressions you made in Activity 2.

Answer. Model sentences: *We had to stop the car — the engine light came on. / She pressed the light switch, and the whole room lit up.* Each sentence gives the meaning of the compound without naming it, so a partner can guess the compound noun.

Compound nouns, Activity 5. Choose any article in a magazine or newspaper and write down all the compound nouns you find.

Answer. In a typical technology or energy article you should expect to find compounds such as power station, solar panel, wind farm, battery pack or energy crisis — list every two-noun combination you meet, and check in a dictionary which ones are written as one word, two words or with a hyphen.

Compound nouns, Activity 6. Look at the compound expressions you have worked with in this unit. Mark all those that you feel you need to be able to use yourself rather than just to understand when others use them.

Answer. A self-study decision: mark for active use the compounds that belong to your own field — for an energy engineering student, for example, expressions such as power station or solar panel — and leave for recognition the ones you only meet in general reading.

4.2.5 Conversion — practice activities (with full answer keys)

Activity 1 — Same word, different job. Write N (noun), V (verb) or Adj (adjective) for each word in bold.

- Can I phone you tonight? ___
- My phone is broken. ___
- We went for a long walk by the river. ___
- I walk to school every day. ___

- Please clean the kitchen. ___
- The kitchen is very clean. ___
- Don't forget to email me the photos. ___
- I got an email from my teacher. ___

Conversion, Activity 1 — answer key (8 items).

Answer. (1) V — phone follows can, so it is a verb; (2) N — my phone; (3) N — a long walk; (4) V — I walk; (5) V — please clean; (6) Adj — very clean; (7) V — to email me; (8) N — an email.

Activity 2 — Noun or adjective to verb. Use the word in brackets as a verb. Change its form if you need to.

- Anna sent me a text. → Anna ___ me this morning. (text)
- I use a brush for my hair. → I ___ my hair every morning. (brush)
- I'm reading a good book. → We have ___ a table for eight. (book)
- She put butter on the toast. → She ___ her toast and sat down. (butter)
- The plants need water. → Please ___ the plants while I'm away. (water)
- The box is empty. → Please ___ the box before you throw it away. (empty)

Conversion, Activity 2 — answer key (6 items).

Answer. (1) Anna texted me this morning. (2) I brush my hair every morning. (3) We have booked a table for eight. (4) She buttered her toast and sat down. (5) Please water the plants while I'm away. (6) Please empty the box before you throw it away.

Activity 3 — Verb to noun. Complete each sentence with a word from the box. Use it as a noun. You can use a word twice. Box: run · drink · smile · help · visit · call · swim.

- I need some ___ with my homework.
- He gave me a big ___ when he saw me.
- Would you like a ___ of water?
- We had a lovely ___ from our grandparents last weekend.
- I'll give you a ___ when I get home.
- After the long ___ in the sea, we were tired.
- She goes for a ___ in the park every morning.

Conversion, Activity 3 — answer key (7 items).

Answer. (1) help; (2) smile; (3) drink; (4) visit; (5) call; (6) swim; (7) run.

Activity 4 — Say it with the right stress. Read each sentence aloud. Mark the stressed syllable of the bold word in capital letters, for example REcord or reCORD.

- This is my favourite record. ___
- Please record the lesson. ___
- We bought fresh produce at the market. ___
- Japan will produce more cars next year. ___

Conversion, Activity 4 — answer key (4 items).

Answer. (1) This is my favourite REcord (noun — stress on the first syllable); (2) Please reCORD the lesson (verb — stress on the second syllable); (3) We bought fresh PROduce at the market (noun); (4) Japan will proDUCE more cars next year (verb).

Activity 5 — New verbs and your own sentences.

Conversion, Activity 5, part 1. Guess the meaning. What does the verb in bold mean? • I googled the restaurant before we went. • We zoomed with our cousins in Canada. • He ghosted me. He never answered my messages. • Let's bookmark this page so we can find it later.

Answer. googled = searched for information about it online (using Google); zoomed = had a video call (using Zoom); ghosted = stopped answering me and ended contact suddenly; bookmarked = saved the page so we can find it again quickly.

Conversion, Activity 5, part 2. Write your own. For each word, write two sentences: one with a noun and one with a verb (plan · cook · wish · laugh).

Answer. Models: What's the plan for tonight? (noun) / We plan to leave early. (verb) · My mum is a very good cook. (noun) / I cook dinner every evening. (verb) · He made a wish before blowing out the candles. (noun) / I wish you good luck. (verb) · She has a lovely laugh. (noun) / Don't laugh at me. (verb).

Conversion, Activity 5, part 3. Talk with a partner. Do you use verbs made from nouns in your own language? Which new ones have you heard recently?

Answer. Open discussion. Model: yes, my own language also turns nouns into verbs, and English keeps producing new ones — the newest conversions in the lecture's watch-out list are to google (search online), to zoom (have a video call), to ghost (stop answering someone) and to bookmark (save a page).

4.2.6 Parts 2 and 3 — reading comprehension and Switch Games

Part 2: Reading comprehension — the reading pages of the Week 7 file are image-based and carry no readable text. Part 3: Oral Expression — Switch Games: the lecture links to the interactive Switch Games activity file (file:///E:/Week%207%20files/Switch%20Games.html); the game is played in class as oral expression practice with the week's word formation material.

5 Appendix

5.1 Passive quick reference

[Source: Week 5 Grammar Worksheet rules page; Week 5 Lecture Language Reference]

Form	Structure	Example
present simple passive	am/is/are + past participle	Oil and gas are found beneath the ground or sea.
present continuous passive	am/is/are being + past participle	Two men are being questioned by the police.
past simple passive	was/were + past participle	Aztec boys were trained to fight.
present perfect passive	has/have been + past participle	Some countries have been harmed by tourism.
will passive	will be + past participle	A new class representative will be chosen on Monday.
agent	by + person, only when really necessary	The new concert hall will be opened by the King. / The two men were arrested. (by the police — this is obvious)
instrument	with + object that performs the action	The game was played with a ball of solid rubber.

Only transitive verbs (verbs with an object) can be made passive; negatives and questions are formed exactly as in active sentences.

5.2 Graph language bank

[Source: Week 6 Lecture, Part 1 (Language help) and Part 2 (cause and effect)]

Verb + preposition patterns from the Week 6 files.

Word	Pattern	Example
increase, decrease (verbs)	+ by (the size of the change)	The population of the city has increased by 10%.
increase, rise, growth, fall, drop, decline, decrease (nouns)	+ of (the size) / + in (what is changing)	a rise / an increase of 15% in the number of cars
effect, impact	+ on	The effect/impact of the film on the audience was very powerful.
reason	+ for / why	What was the reason for his success / why he succeeded?
precedent	+ for	There are several precedents for taking such a decision.
contribute	+ to	Mobile technology contributed to the information revolution.
stem, spring	+ from	Sid's determination stems from his desire to improve the world.
account	+ for	The tilting of the earth on its axis accounts for the change in the seasons.
give rise	+ to	The country's victory gave rise to a new mood in society.
derive	+ from	A number of benefits can be derived from this situation.
influence (verb)	+ object, no preposition	Her grandmother influenced Sarah's choice of career.
reach a peak / exceed / fall below	—	The amount received reaches a peak at the age of 18; exceed [another number] — opposite = fall below.

5.3 Noun formation table

[Source: Week 7 Lecture, Part 1 — word formation techniques]

Suffix	Added to	Examples
-tion	verbs	act → action; inspect → inspection; produce → production
-ment	verbs	develop → development; measure → measurement
-ness	adjectives	happy → happiness; dark → darkness
-ity	adjectives	able → ability; possible → possibility
-er	verbs	build → builder; design → designer
-or	verbs	act → actor; generate → generator

Course-attested noun formations from the Week 6 cause-and-effect strand: motivate → motivation; influence → influence; trigger → trigger; contribute → contribution; induce → inducement. Conversion reminder: water (noun) → to water (verb) adds nothing at all («zero derivation»).

5.4 Ten one-line answers

Revision 1. How do you form the present simple passive?

Answer. With am/is/are + past participle — This room isn't used very often.

Revision 2. How do you form the past simple passive?

Answer. With was/were + past participle — The college was built in the 16th century.

Revision 3. When do you use by in a passive sentence?

Answer. When it is important to know the agent — Some countries have been harmed by tourism; otherwise leave the agent out.

Revision 4. When do you use with?

Answer. For objects which perform actions — The game was played with a ball of solid rubber.

Revision 5. Which verbs can be made passive?

Answer. Only transitive verbs — verbs that have an object (I lost my wallet, not I swim every day).

Revision 6. Why is the passive the standard form for technical processes?

Answer. Because the steps matter, not the doer — Oil and gas are found beneath the ground or sea; a hole is made using a drill.

Revision 7. Which sentence is correct: «an increase in 12% of smart phone sales» or «an increase of 12% in smart phone sales»?

Answer. An increase of 12% in smart phone sales — of for the size of the change, in for what is rising.

Revision 8. What do two lines on a graph do if they intersect, and if they run parallel?

Answer. Intersecting lines cross at a point; parallel lines never intersect.

Revision 9. Name three noun-forming suffixes with an example each.

Answer. -tion (action), -ment (development), -ness (happiness) — also -ity (ability), -er (builder), -or (actor).

Revision 10. What is conversion?

Answer. A word changes its job without changing its shape — water (noun) → to water (verb); teachers call it «zero derivation».